

Public Available Specification

CiA 401-2

Version 3.1.0

CANopen profile for I/O devices

Part 2: Joysticks

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HISTORY

Date	Changes
2013-02-12	<i>Publication of Version 3.1</i> as draft standard proposal (The CiA 401 specification has been split into two parts) - <i>TPDO 1 to 2 and RPDO 1 parameter sets added</i> NOTE: This document has been converted into “docx format”. The conversion caused minor layout differences to the predecessor document in “doc format”. The technical content word-by-word is the very same.

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1 Scope

This CANopen profile specifies generic CANopen digital and analog input and output devices. Devices compliant to this specifications use communication techniques, which conform to those described in the CANopen application layer and communication profile specification. In addition, programmable I/O devices may use communication techniques, which conform to those described in the CANopen additional application layer functions.

This profile consists of several parts:

- Part 1: Generic I/O devices
- Part 2: Joysticks

This part of the profile specifies the CANopen interface for joysticks and similar devices, e.g. foot-pedals.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

/CiA301/	CiA 301, CANopen application layer and communication profile
/CiA303-1/	CiA 303-1, CANopen Recommendation – Part 1: Cabling and connector pin assignment
/CiA303-2/	CiA 303-2, CANopen Recommendation – Part 2: Representation of SI units and prefixes
/CiA305/	CiA 305, CANopen layer setting services
/CiA401-1/	CiA 401-1, CANopen device profile for generic I/O devices

3 Terms and definitions

For the purpose of this document, the following terms and definitions and those given in /CiA301/, /CiA303-1/, /CiA303-2/, /CiA305/ and /CiA401-1/ apply.

4 Symbols and abbreviated terms

For the purpose of this document, the following symbols and abbreviated terms and those given in /CiA301/, /CiA303-1/, /CiA303-2/, /CiA305/ and /CiA401-1/ apply.

5 Operating principles

5.1 General

Devices compliant to this device profile support digital inputs, analog inputs and optionally digital outputs. A single joystick may have 1-, 2-, or 3-axes of movement. The joystick with 3-axes may support a rotational movement around one axis, additionally, to the movements along each of two axes as for example forward to backward and left to right movements.

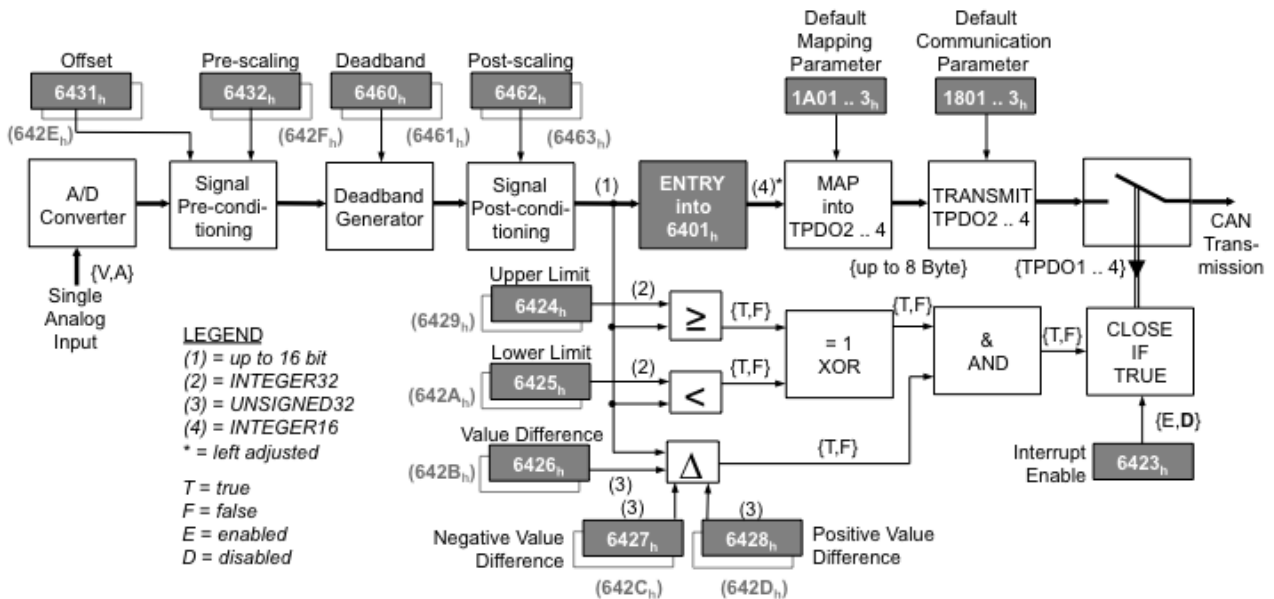
NOTE 1 The analog inputs are the proportional joystick input values, the digital inputs are the buttons for memory functions respectively additional functions and the digital outputs are for indicating LEDs or general purposes.

NOTE 2 A joystick device that provides two independent joystick functions (a base joystick operated by the hand and a second one on top of the other operated by the thumb) implements a second logical CANopen device (for details see /CiA301/) or uses the next following input functions in the object dictionary.

5.2 Joystick signal conditioning

The analog inputs representing the x, y, and z dimension are pre-scaled and shifted with the offset by using the objects 6432_h (642F_h) and 6431_h (642E_h), respectively (see /CiA401-1/). If

a dead-band and post-scaling is required, the objects 6460_h (6461_h) and 6462_h (6463_h) respectively are used. The configuration of the analog input dead-band float object (6461_h) causes a value change in the object 6460_h. On the other hand, the configuration of the analog input dead-band unsigned object 6460_h causes a value change in the object (6461_h). The configuration of the analog input post-scaling float object (6463_h) causes a value change in the object 6462_h. On the other hand, the configuration of the analog input post-scaling unsigned object (6462_h) causes a value change in the object (6463_h). Figure 1 specifies the relation between the analog input objects with pre-scaling, dead-band and post-scaling functionality for 16-bit and 32-bit access.



NOTE The parameters representing REAL32 values are put in the brackets. The parameters representing INTEGER32 values are not put in the brackets.

Figure 1 – Analog Input with dead-band-functionality

5.3 Joystick proportional inputs

The proportional inputs representing the joystick position use the functionality for analog inputs with signed 16-bit access.

NOTE Joystick devices provide generally two or three 16-bit analog position values representing the x-, y- respectively z-dimension.

5.4 Joystick memory function

The memory buttons for x-, y-, and z-axis may freeze or release the proportional values of the related axis.

5.5 Foot pedal

Foot pedals are represented as a joystick with 1-axis of movement, which provides button functionality as well as position values for movements along one axis. Foot pedal groups may be interpreted as multi-axis joysticks or multiple multi-axis joysticks.

6 Physical layer specification

6.1 General

This clause specifies the physical layer used in this device profile.

6.2 Transmission rates

The device compliant to this device profile shall support one or more bit-rates as specified in /CiA301/. The bit-rate assignment by means of LSS (see /CiA305/) shall be supported. An additional bit-rate assignment method is manufacturer-specific. The specification of the automatic bit rate detection is not in the scope of this specification.

6.3 Connectors

The pin-assignment of the used connectors shall comply with the recommendations given in /CiA303-1/.

NOTE It is recommended to use a pair of connectors of the same connector type. The male-female connector pair is recommended as long as supported by used connector type.

7 Data link layer

The data link layer shall comply with /CiA303-1/.

8 Application layer

8.1 General

This clause provides an overview on the application layer functionality and specifies, in general, the communication behavior.

8.2 Node-ID assignment

The node-ID assignment by means of LSS shall be supported. Additional node-ID assignment methods are manufacturer-specific.

NOTE The node-ID assignment via the CANopen object dictionary is not recommended. Configuring the node-ID by means of SDO can lead to serious problems, when using the store and restore functions as specified in /CiA305/.

8.3 Network management and Heartbeat functionality

Devices compliant with this device profile shall support NMT slave functionality. Node and or Life Guarding are no longer recommended, but may be supported (NOTE).

NOTE Heartbeat producer is highly recommended in the new implementations.

8.4 EMCY functionality

The device shall support the production of EMCY messages. It is recommended to support all Emergency Error Codes defined in /CiA301/ and /CiA401-1/, if appropriate. If the joystick device transits to NMT operational state and the *analog input global interrupt* object (6423_h) is set to FALSE, it shall transmit an Emergency message with the error code 0080_h. This Emergency message shall not cause a local transition into *NMT pre-operational* or *NMT stopped* state.

9 Device classes

There are no device classes specified.

10 General communication parameter

10.1 General

The general communication parameter are specified in /CiA301/. This clause provides additional value specifications or specifies some optional parameters as mandatory.

10.2 Object 1000_h: Device type

This object indicates the type and functionality of the device. The structure as well as the object and entry descriptions are defined in /CiA301/; the *device profile number* field as well as the *I/O functionality* and *M* field are specified in /CiA401-1/. Table 1 specifies the values to be used for joysticks and similar devices.

Table 1 — Value definitions

Additional information			Device profile number	Description
Specific functionality	M	I/O functionality		
0000 0001 _b	0 _b	0101 _b	0191 _h	Joystick with digital inputs without digital outputs
0000 0001 _b	0 _b	0111 _b	0191 _h	Joystick with digital inputs and digital outputs
0000 0010 _b	1 _b	011X _b	0191 _h	3-D joystick with special PDO mapping
0000 0011 _b	1 _b	011X _b	0191 _h	2-D joystick with special PDO mapping
X = don't care				

10.3 Object 1001_h: Error register

See /CiA401-1/.

10.4 Object 1014_h: Emergency

The device may support the production of EMCY messages.

10.5 Object 1016_h: Consumer heartbeat time

Heartbeat consumer functionality may be conditionally supported (NOTE).

NOTE In case the device supports output functions.

10.6 Object 1017_h: Producer heartbeat time

Heartbeat producer shall be supported in the new implementations.

10.7 Object 1018_h: Identity

The device manufacturer shall assign uniquely the product code, the revision number, and serial number.

NOTE The vendor-ID is assigned by CAN in Automation. CiA recommends supporting all four sub-indexes of this parameter.

10.8 Object 1029_h: Error behavior

If this object is not implemented, the device shall behave as specified by the default values specified in /CiA401-1/.

11 PDO parameter sets**11.1 General**

The joystick may support a generic mapping for generic joysticks or a specific mapping for 2-D or 3-D joystick.

NOTE If additional proportional inputs are required, TPDOs 3 and 4 are used (see /CiA401-1/).

11.2 TPDO 1

The 1st TPDO transmits the values of maximum 8 x 8 buttons. Table 2 specifies the object description and Table 3 specifies the entry description of the TPDO communication parameter. Table 4 specifies the object description and Table 5 specifies the entry description of the TPDO mapping parameter. In case of 3-D joysticks (if the specific functionality field is 02_h, see object 1000_h), the joystick shall use the first TPDO with the mapping as specified in Table 6. In case of 2-D joysticks (if the specific functionality field is 03_h, see object 1000_h), the joystick shall use the first TPDO with the mapping as specified in Table 7.

NOTE The number of mapped objects into this TPDO depends on the hardware. The analogue input setup objects are used in order to transmit only the first proportional value different from 0.

Table 2 — Object description

Attribute	Value
INDEX	1800 _h
Name	TPDO 1 communication parameter
Object code	RECORD
Data type	PDO communication parameter record
Category	Conditional: Mandatory, if the specific functionality field is 01 _h , 02 _h or 03 _h (see object 1000 _h)

Table 3 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	02 _h to 06 _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	COB-ID used by TPDO
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	{0000 0180 _h , 4000 0180 _h } + node-ID
Sub-index	02 _h
Description	Transmission type
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	255 _d
Sub-index	03 _h
Description	Inhibit time
Entry category	Optional
Access	rw
PDO mapping	No
Value range	See /CiA301/
Default value	00 _h

Attribute	Value
Sub-index	05 _h
Description	Event timer
Entry category	Optional
Access	rw
PDO mapping	No
Value range	See /CiA301/
Default value	00 _h , unequal 0 if periodic transmission supported
Sub-index	06 _h
Description	SYNC start value
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	UNSIGNED8
Default value	See /CiA301/

Table 4 — Object description

Attribute	Value
INDEX	1A00 _h
Name	TPDO 1 mapping parameter
Object code	RECORD
Data type	PDO mapping parameter record
Category	Conditional: Mandatory, if 1800 _h is implemented

Table 5 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Number of mapped application objects in TPDO
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	Manufacturer-specific (01 _h to 08 _h)
Sub-index	01 _h
Description	1 st application object
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6000 01 08 _h

Attribute	Value
Sub-index	02 _h
Description	2 nd application object
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6000 02 08 _h
to	
Sub-index	08 _h
Description	8 th application object
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6000 08 08 _h

Table 6 – Specific TPDO 1 mapping for 3-D joysticks

Mapping entry	Index and sub-index	Description
1	6401 _h 01 _h	Proportional value in x direction
2	6401 _h 02 _h	Proportional value in y direction
3	6401 _h 03 _h	Proportional value in z direction
4	6000 _h 02 _h	Additional buttons
5	6000 _h 01 _h	Memory buttons for x-, y-, z-axis

Table 7 – Specific TPDO 1 mapping for 2-D joysticks

Mapping entry	Index and sub-index	Description
1	6401 _h 01 _h	Proportional value in x direction
2	6401 _h 02 _h	Proportional value in y direction
3	6000 _h 04 _h	Additional buttons
4	6000 _h 03 _h	Additional buttons
5	6000 _h 02 _h	Additional buttons
6	6000 _h 01 _h	Memory buttons for x-, y-axis

11.3 TPDO 2

The 2nd TPDO transmits the 16-bit values of maximum 4 proportional inputs. Table 8 specifies the object description and Table 9 specifies the entry description of the TPDO communication parameter. Table 10 specifies the object description and Table 11 specifies the entry description of the TPDO mapping parameter.

NOTE The number of mapped objects into this TPDO depends on the hardware.

Table 8 — Object description

Attribute	Value
INDEX	1801 _h
Name	TPDO 2 communication parameter
Object code	RECORD
Data type	PDO communication parameter record
Category	Conditional: Mandatory, if the specific functionality field is 01 _h (see object 1000 _h)

Table 9 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	02 _h to 06 _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	COB-ID used by TPDO
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	{0000 0280 _h , 4000 0280 _h } + node-ID
Sub-index	02 _h
Description	Transmission type
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	255 _d
Sub-index	03 _h
Description	Inhibit time
Entry category	Optional
Access	rw
PDO mapping	No
Value range	See /CiA301/
Default value	00 _h

Attribute	Value
Sub-index	05 _h
Description	Event timer
Entry category	Optional
Access	rw
PDO mapping	No
Value range	See /CiA301/
Default value	00 _h , unequal 0 if periodic transmission supported
Sub-index	06 _h
Description	SYNC start value
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	UNSIGNED8
Default value	See /CiA301/

Table 10 — Object description

Attribute	Value
INDEX	1A01 _h
Name	TPDO 2 mapping parameter
Object code	RECORD
Data type	PDO mapping parameter record
Category	Conditional: Mandatory, if 1801 _h is implemented

Table 11 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Number of mapped application objects in TPDO
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	Manufacturer-specific (01 _h to 04 _h)
Sub-index	01 _h
Description	1 st application object
Entry category	Mandatory
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6401 01 10 _h

Attribute	Value
Sub-index	02 _h
Description	2 st application object
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6401 01 10 _h
to	
Sub-index	04 _h
Description	4 th application object
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	Manufacturer-specific analog input

11.4 RPDO 1

This RPDO receives the values of up to 64 digital outputs in joystick (LED or general purpose). Table 12 specifies the object description and Table 13 specifies the entry description of the RPDO communication parameter. Table 14 specifies the object description and Table 15 specifies the entry description of the RPDO mapping parameter.

NOTE The number of mapped objects into this RPDO depends on the hardware.

Table 12 — Object description

Attribute	Value
INDEX	1400 _h
Name	RPDO 1 communication parameter
Object code	RECORD
Data type	PDO communication parameter record
Category	Conditional; Mandatory, if digital outputs are supported (see object 1000 _h)

Table 13 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	02 _h
Default value	02 _h

Attribute	Value
Sub-index	01 _h
Description	COB-ID used by RPDO
Entry category	Mandatory
Access	<i>rw or const</i>
PDO mapping	No
Value range	See /CiA301/
Default value	0000 0200 _h + node-ID
Sub-index	02 _h
Description	Transmission type
Entry category	Mandatory
Access	<i>rw or const</i>
PDO mapping	No
Value range	See /CiA301/
Default value	255 _d

Table 14 — Object description

Attribute	Value
INDEX	1600 _h
Name	RPDO 1 mapping parameter
Object code	RECORD
Data type	PDO mapping parameter record
Category	Conditional: Mandatory, if 1400 _h is implemented

Table 15 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Number of mapped application objects in RPDO
Entry category	Mandatory
Access	<i>rw or const</i>
PDO mapping	No
Value range	See /CiA301/
Default value	Manufacturer-specific (01 _h to 08 _h)
Sub-index	01 _h
Description	1 st application object
Entry category	Mandatory
Access	<i>rw or const</i>
PDO mapping	No
Value range	See /CiA301/
Default value	6200 01 08 _h

Attribute	Value
Sub-index	02 _h
Description	2 nd application object
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6200 02 08 _h
to	
Sub-index	08 _h
Description	8 th application object
Entry category	Optional
Access	rw or const
PDO mapping	No
Value range	See /CiA301/
Default value	6200 08 08 _h

12 Detailed application parameter specification

12.1 General

Digital and analog parameters (see /CiA401-1/) used for implementation in joysticks, are specified in this clause (NOTE). The list of the application objects used for the implementation of the joysticks and similar devices is provided in the Annex A.

NOTE The value structure of these parameters is adapted for use in joystick implementation.

12.2 Object 6000_h: Read 8-bit

This object shall be supported. The device shall use the first byte of the digital input functionality (see object 6000_h 01_h in /CiA401-1/) for the joystick memory function for x-, y-, and z-axis. If the joystick does not support a memory function, the memory function bit, assigned to the unsupported axe, shall be equal 0. The joystick may additionally support up to 24 buttons. The value structure of joystick memory function for x-, y-, and z-axis is specified in Figure 2. The value structure of additional buttons (sub-indexes 02_h to 04_h) is specified in Figure 3. Table 16 to Table 19 specifies the value definition for sub-indexes 01_h to 04_h.

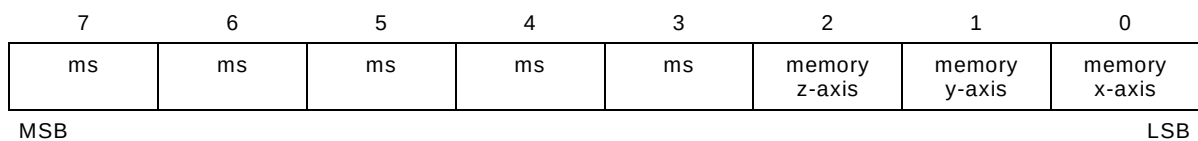


Figure 2 — Value structure of sub-index 01_h

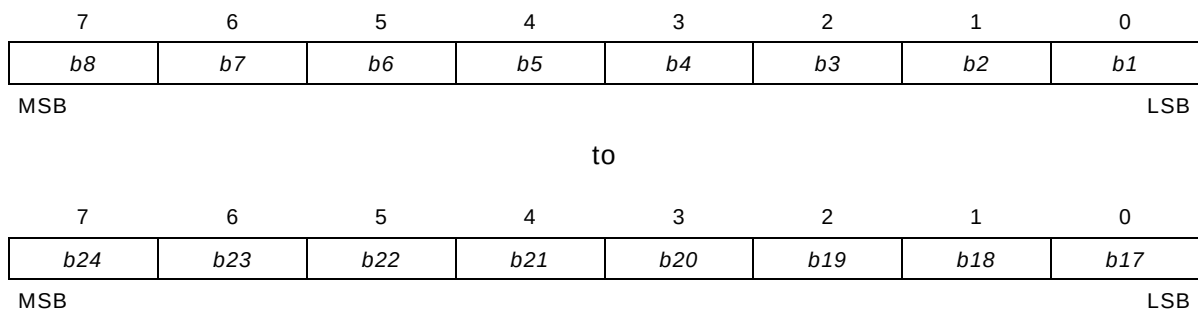


Figure 3 – Value structure for sub-index 02_h to 04_h

Table 16 – Value definition for sub-index 01_h

Name	Bit	Value	Description
memory x-axis	0	0 _b	Memory function off (NOTE 1)
		1 _b	Memory function on (NOTE 2)
memory y-axis	1	0 _b	Memory function off (NOTE 1)
		1 _b	Memory function on (NOTE 2)
memory z-axis	2	0 _b	Memory function off (NOTE 1)
		1 _b	Memory function on (NOTE 2)
ms: Manufacturer-specific	3 to 7	0 _b	Manufacturer-specific function
		1 _b	

NOTE 1 Release proportional value of related axis.

NOTE 2 Freeze proportional value of related axis.

NOTE 3 It is optional to include the freezing function into the joystick itself. In this case, the proportional values are not modified any more if the memory function is switched on. Otherwise, the freezing is done by the application and the buttons are interpreted as freezing request. In this case, the memory buttons have no effect on the proportional input values.

Table 17 – Value definition for sub-index 02_h

Name	Bit	Value	Description
b1: button	0	0 _b	Off
		1 _b	On
to			
b8: button	7	0 _b	Off
		1 _b	On

Table 18 – Value definition for sub-index 03_h

Name	Bit	Value	Description
b9: button	0	0 _b	Off
		1 _b	On
to			
b16: button	7	0 _b	Off
		1 _b	On

Table 19 – Value definition for sub-index 04_h

Name	Bit	Value	Description
b17: button	0	0 _b	Off
		1 _b	On
to			
b24: button	7	0 _b	Off
		1 _b	On

12.3 Object 6401_h: Read analogue input 16-bit

This object (see /CiA401-1/) shall be supported and all related configuration objects may be implemented.

12.4 Object 6430_h: Analogue input SI unit

This object may be supported (see /CiA401-1/). Sub-index 01_h corresponds to the x-axis, sub-index 02_h corresponds to the y-axis and sub-index 03_h corresponds to the z-axis. Table 20 specifies the entry description.

NOTE An analog value of zero represents the zero position of the joystick.

Table 20 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	01 _h to FE _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	Analog input 1
Entry category	Mandatory
Access	rw
PDO mapping	Default (see PDO mapping)
Value range	UNSIGNED32
Default value	0000 _h
Sub-index	02 _h
Description	Analog input 2
Entry category	Optional
Access	rw
PDO mapping	Default (see PDO mapping)
Value range	UNSIGNED32
Default value	0000 _h
to	
Sub-index	04 _h
Description	Analog input 4
Entry category	Optional
Access	rw
PDO mapping	Default (see PDO mapping)
Value range	UNSIGNED32
Default value	0000 _h

12.5 Object 6460_h: Analog input dead-band unsigned

This object shall set the dead-band for the analog inputs (6401 01_h to 03_h). A configuration of the analog input dead-band float object (6461_h) shall also cause a value change in the 6460_h object and vice versa. Table 21 specifies the object description and Table 22 specifies the entry description.

Table 21 — Object description

Attribute	Value
INDEX	6460 _h
Name	Analog input dead-band unsigned
Object code	ARRAY
Data type	UNSIGNED32
Category	Optional

Table 22 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	03 _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	Analog input 01 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	UNSIGNED32
Default value	0000 0000 _h
Sub-index	02 _h
Description	Analog input 02 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	UNSIGNED32
Default value	0000 0000 _h
Sub-index	03 _h
Description	Analog input 03 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	UNSIGNED32
Default value	0000 0000 _h

12.6 Object 6461_h: Analog input dead-band float

This object shall set the dead-band for the analog inputs (6401 01_h to 03_h). A configuration of the analog input dead-band unsigned object (6460_h) shall also cause a value change in the

6461_h object and vice versa. Table 23 specifies the object description and Table 24 specifies the entry description.

Table 23 — Object description

Attribute	Value
INDEX	642A _h
Name	Analog input dead-band float
Object code	ARRAY
Data type	REAL32
Category	Optional

Table 24 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	03 _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	Analog input 01 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	REAL32
Default value	0,0
Sub-index	02 _h
Description	Analog input 02 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	REAL32
Default value	0,0
Sub-index	03 _h
Description	Analog input 03 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	REAL32
Default value	0,0

12.7 Object 6462_h: Analog input post-scaling unsigned

This object shall set the gain after the addition of the dead-band-functionality for the analog inputs (6401 01_h to 03_h). Configuration of the analog input post-scaling float object (6463_h) shall also cause a value change in the 6462_h object and vice versa. Table 21 specifies the object description and Table 22 specifies the entry description.

Table 25 — Object description

Attribute	Value
INDEX	6462 _h
Name	Analog input post-scaling unsigned
Object code	ARRAY
Data type	UNSIGNED32
Category	Optional

Table 26 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	01 _h to 03 _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	Analog input 01 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	UNSIGNED32
Default value	0000 0000 _h
Sub-index	02 _h
Description	Analog input 02 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	UNSIGNED32
Default value	0000 0000 _h

Attribute	Value
Sub-index	03 _h
Description	Analog input 03 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	UNSIGNED32
Default value	0000 0000 _h

12.8 Object 6463_h: Analog input post-scaling float

This object shall set the gain after the addition of the dead-band-functionality for the analog inputs (6401 01_h to 03_h). Configuration of the analog input post-scaling unsigned object (6462_h) shall also cause a value change in the 6463_h object and vice versa. Table 27 specifies the object description and Table 28 specifies the entry description.

Table 27 — Object description

Attribute	Value
INDEX	6463 _h
Name	Analog input post-scaling float
Object code	ARRAY
Data type	REAL32
Category	Optional

Table 28 — Entry description

Attribute	Value
Sub-index	00 _h
Description	Highest sub-index supported
Entry category	Mandatory
Access	const
PDO mapping	No
Value range	01 _h to 03 _h
Default value	Manufacturer-specific
Sub-index	01 _h
Description	Analog input 01 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	REAL32
Default value	0,0

Attribute	Value
Sub-index	02 _h
Description	Analog input 02 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	REAL32
Default value	0,0
Sub-index	03 _h
Description	Analog input 03 _h
Entry category	Mandatory
Access	rw
PDO mapping	Optional
Value range	REAL32
Default value	0,0

Annex A Communication and application parameters

The communication parameters are specified in Table 29. The application parameters for 16-bit access (see clause 5.2) are specified in the Table 30.

Table 29 – Communication parameters

Index	Object name	Category	Reference
1005 _h	COB-ID SYNC	O	/CiA301/
1007 _h	Synchronous window length	O	/CiA301/
1010 _h	Store parameters	O	/CiA301/
1011 _h	Restore default parameters	O	/CiA301/
1014 _h	COB-ID EMCY	O	/CiA301/
1015 _h	Inhibit time EMCY	O	/CiA301/
1017 _h	Producer heartbeat time	O	/CiA301/
1029 _h	Error behavior	O	/CiA301/, /CiA401-1/

Table 30 – Application parameters (dead band functionality for 16-bit access)

Index	Object name	Category	Reference
6000 _h	Read input 8-bit	M	/CiA401-1/
6200 _h	Write output 8-bit	O	/CiA401-1/
6401 _h	Read analogue input 16-bit	M	/CiA401-1/
6423 _h	Analogue input global interrupt enable	O	/CiA401-1/
6424 _h	Analogue input interrupt upper limit integer	O	/CiA401-1/
6425 _h	Analogue input interrupt lower limit integer	O	/CiA401-1/
6426 _h	Analogue input interrupt delta unsigned	O	/CiA401-1/
6427 _h	Analogue input interrupt negative delta unsigned	O	/CiA401-1/
6428 _h	Analogue input positive delta unsigned	O	/CiA401-1/
6430 _h	Analogue input SI unit	O	/CiA401-1/
6431 _h	Analogue input offset integer	O	/CiA401-1/
6432 _h	Analogue input pre-scaling integer	O	/CiA401-1/
6460 _h	Analogue input dead-band unsigned	O	/CiA401-2/
6462 _h	Analogue input post-scaling unsigned	O	/CiA401-2/

Table 31 – Application parameters (dead band functionality for 32-bit access)

Index	Object name	Category	Reference
6000 _h	Read input 8-bit	M	/CiA401-1/
6200 _h	Write output 8-bit	O	/CiA401-1/
6401 _h	Read analogue input 16-bit	O	/CiA401-1/
6423 _h	Analogue input global interrupt enable	O	/CiA401-1/
6429 _h	Analogue input interrupt upper limit float	O	/CiA401-1/
642A _h	Analogue input interrupt lower limit float	O	/CiA401-1/
642B _h	Analogue input interrupt delta float	O	/CiA401-1/
642C _h	Analogue input interrupt negative delta float	O	/CiA401-1/
642D _h	Analogue input positive delta float	O	/CiA401-1/
642E _h	Analogue input offset float	O	/CiA401-1/
642F _h	Analogue input pre-scaling float	O	/CiA401-1/
6430 _h	Analogue input SI unit	O	/CiA401-1/
6461 _h	Analogue input dead-band float	O	/CiA401-2/
6463 _h	Analogue input post-scaling float	O	/CiA401-2/